Ultra-Low-Resolution Thermal Sensing Can Estimate Walkable Directions Without Calibration or Manual Labels

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Motivation: Wearable navigation still depends on detailed vision, calibration, and manual labels

- Wearable navigation aids rely on high-resolution cameras and depth sensors, raising privacy concerns.
- Building these systems requires costly hand-labeled datasets for walkable directions.
- Most systems need geometric calibration and a fixed camera layout, which breaks if a sensor shifts.
- **Can ultra-low-resolution thermal sensing alone estimate walkable directions, without calibration or labels?**

Methods: A calibration-free, RGB-supervised framework for ultra-low-resolution thermal sensing

- A wearable rig with three ultra-low-resolution thermal cameras captures multi-view data while walking.
- A high-resolution RGB camera, used only during data collection, auto-generates walkable/obstructed labels via object detection.
- At inference time, the model needs only the thermal cameras: no RGB, calibration, or fixed camera layout.

Finding 1: Feature fusion wins out among the three fusion strategies

- Feature fusion's per-camera backbone passes capture cross-camera detail that channel-stacking's single pass misses.
- Pixel-stitching lags both, since warping blurs the thermal signal before the model ever sees it.
- The extra backbone passes cost more compute, but the accuracy gain on these ultra-low-resolution tiles is worth it.

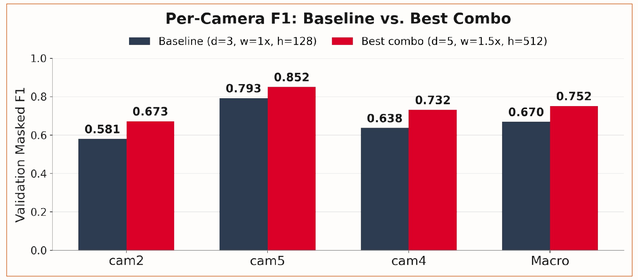


Fig. 1: Preliminary per-camera F1, baseline vs. best fusion combo found so far (to be replaced with a comparison across all three fusion strategies + shuffled experiment performance).

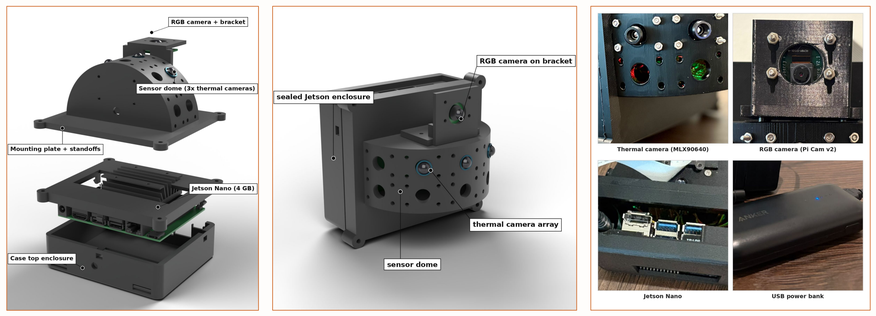


Fig. 2: Exploded rig view, assembled sensor enclosure, and individual components (thermal camera, RGB camera, Jetson nano, USB power bank)

Finding 2: Direction predictions hold steady when camera slots are shuffled

- The model relies on directionally meaningful features, not fixed camera positions: shuffling slots leaves masked F1 largely unchanged.
- Performance holds across shuffles, confirming the model isn't overfit to which physical camera feeds which input channel.
- The identity-tagged model and shuffle wrapper test this without touching the underlying dataset files.

Implications: Calibration-free, ultra-low-res sensing broadens access

- Each thermal camera captures only a 32x24-pixel array, which is too coarse to identify anyone, yet still supports reliable navigation.
- No manual labels and sensor calibration lowers the barrier to entry.
- Contributes new fusion-design and camera-placement robustness research, even at this resolution.

References

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